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THREE-WAY VALVE

Abstract

To assemble a three-way valve without weakening the bond between a driver and a shaft with grease adhering to the shaft end, a straight flow path includes two ends and an intermediate portion respectively continuous with first to third ports, and first and second valve orifices having opening areas changed when first and second valve elements (**34a**, **34b**) at the bottom of the shaft (**35**) move. The shaft extends through a bearing (**42**) in a guide (**41**) with its top bonded inside the cylindrical driver (**50**). The driver includes a gear (**51**) at the top and an externally threaded portion (**53**) at the bottom. The guide includes an internally threaded portion (**43**) above the bearing to receive the externally threaded portion. The driver includes a cylindrical portion (**54**) protruding downward from the externally threaded portion with its outer diameter smaller than an inner diameter of the internally threaded portion.

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Background/Summary

BACKGROUND OF INVENTION

Field of the Invention

[0001] The present invention relates to a three-way valve including a straight flow path having an end continuous with a first port, the other end continuous with a second port, and an intermediate portion continuous with a third port to change the amount of fluid flowing between the first port and the third port and the amount of fluid flowing between the second port and the third port.

Background Art

[0002] A three-way valve including three ports connected to pipes is common in, for example, a water heater. Such a valve may be a flow divider valve for dividing a flow of fluid (e.g., water) in a single pipe into two pipes at a changeable flow ratio between the two pipes, or a mixing valve for combining flows of fluid in two pipes into a single pipe at a changeable mixing ratio. For example, Patent Literature 1 describes a three-way valve used as a flow divider valve in a water heater. The valve includes a casing containing a straight flow path. The straight flow path has an end continuous with a first port, the other end continuous with a second port, and an intermediate portion continuous with a third port. The straight flow path includes a first valve orifice between the first port and the third port. The first valve orifice has an opening area changeable in response to a first valve element moving toward or away from the first valve orifice. Similarly, the straight flow path includes a second valve orifice between the second port and the third port. The second valve orifice has an opening area changeable in response to a second valve element moving toward or away from the second valve orifice. The first valve element and the second valve element are bonded to an end of a shaft that is drivable by a motor to move in the axial direction parallel to the straight flow path. The first valve element and the second valve element move along the straight flow path together with the shaft to change the amount of fluid flowing between the first port and the third port and the amount of fluid flowing between the second port and the third port.

[0003] The shaft is placed through a bearing in an inner member, or a guide, incorporated in the casing of the three-way valve. The shaft is supported by the bearing and is movable in the axial direction. The distal end of the shaft opposite to the first valve element and the second valve element is received inside a driver and bonded to the driver. The driver includes, on its outer peripheral surface, a gear to which a driving force is transmitted from the motor. The driver includes, on its outer peripheral surface, the gear adjacent to the upper end and includes an externally threaded portion with an external thread adjacent to the lower end, when the direction in which the valve elements are located with respect to the shaft is downward and the direction in which the driver is located with respect to the shaft is upward in the axial direction. The inner member includes, upward from the bearing, an internally threaded portion having an inner peripheral surface with an internal thread. The internally threaded portion corresponds to the externally threaded portion and receives the externally threaded portion. When the driver is driven by the motor and rotates about the axis of the shaft, the shaft moves in the axial direction together with the driver in response to the externally threaded portion being screwed into the internally threaded portion.

CITATION LIST

Patent Literature

[0004] Patent Literature 1: Japanese Unexamined Patent Application Publication No. 2014-62559

SUMMARY OF INVENTION

[0005] In the three-way valve with the above structure, grease is typically applied to the externally threaded portion on the driver during assembly to allow the externally threaded portion on the driver to be smoothly screwed into the internally threaded portion on the inner member. The grease

may adhere to the end face of the driver adjacent to the external thread and may further adhere to the distal end of the shaft inserted into the driver. The grease placed between the driver and the shaft may decrease their bonding strength.

[0006] In response to the above issue with the known technique, one or more aspects of the present invention are directed to a three-way valve that can be assembled without a decrease in the bonding strength between a driver and a shaft caused by grease applied to an externally threaded portion on the driver adhering to the distal end of the shaft.

[0007] A three-way valve according to aspects of the present invention has the structure below.

First Aspect

[0008] The three-way valve includes a first port, a second port, a third port, a straight flow path, a shaft, a guide, and a driver. The straight flow path includes an end continuous with the first port, another end continuous with the second port, an intermediate portion continuous with the third port, a first valve orifice, and a second valve orifice. The first valve orifice is between the first port and the third port, and has an opening area changeable in response to a first valve element moving toward or away from the first valve orifice. The second valve orifice is between the second port and the third port, and has an opening area changeable in response to a second valve element moving toward or away from the second valve orifice. The shaft is drivable by a motor to move in an axial direction of the shaft parallel to the straight flow path. The shaft includes a first end at which the first valve element and the second valve element are located.

[0009] The first valve element and the second valve element are movable along the straight flow path together with the shaft to change an amount of fluid flowing between the first port and the third port and an amount of fluid flowing between the second port and the third port. The guide includes a bearing through which the shaft is placed. The bearing supports the shaft to allow the shaft to move in the axial direction. The driver is cylindrical and includes, on an outer peripheral surface of the driver, a gear to which a driving force is transmitted from the motor. The driver has an insertion hole being open in an end face of the driver facing the first valve element and the second valve element. The insertion hole receives a second end of the shaft opposite to the first valve element and the second valve element. The driver is bonded to the second end of the shaft. Where a direction in which the driver is located with respect to the shaft in the axial direction is upward, and an opposite direction in which the first valve element and the second valve element are located with respect to the shaft in the axial direction is downward, the driver includes, downward from the gear, an externally threaded portion having an outer peripheral surface with an external thread, and the guide includes, upward from the bearing, an internally threaded portion having an inner peripheral surface with an internal thread.

[0010] When the driver is driven by the motor and rotates about an axis of the shaft with the externally threaded portion placed in the internally threaded portion, the shaft moves in the axial direction in response to the externally threaded portion being screwed into the internally threaded portion. The driver includes a cylindrical portion protruding downward from a lower end face of the externally threaded portion. The cylindrical portion includes the insertion hole inside the cylindrical portion. The cylindrical portion has an outer diameter smaller than an inner diameter of the internally threaded portion.

[0011] The three-way valve according to the first aspect includes the cylindrical portion between the externally threaded portion on the outer peripheral surface of the driver and the lower end face (the end face facing the first valve element and the second valve element) of the driver in which the insertion hole is open. In this structure, the grease applied to the externally threaded portion is less likely to adhere to a portion around the insertion hole in the lower end face of the driver (cylindrical portion) than in a structure including no cylindrical portion. The grease is thus less likely to adhere to the distal end of the shaft inserted into the insertion hole in the driver. This can avoid a decrease in the bonding strength between the driver and the shaft caused by the grease.

Second Aspect

[0012] In the three-way valve according to the first aspect, the shaft includes a rotation locking portion having a rotation locking shape being a noncircle in a cross section perpendicular to the axial direction. The driver includes a fitting portion inside the cylindrical portion. The fitting portion is a part of the insertion hole. The fitting portion has a shape corresponding to the rotation locking shape to fit on the rotation locking portion. The rotation locking portion has an upper end located downward from an upper end of the internally threaded portion when the shaft placed through the bearing in the guide is at an uppermost position to which the shaft is movable in the axial direction. The cylindrical portion in the driver protrudes by a length allowing the fitting portion to fit on the rotation locking portion before the externally threaded portion is screwed into the internally threaded portion.

[0013] In the three-way valve according to the second aspect, the driver with the cylindrical portion placed inside the internally threaded portion can be rotated freely about the axis of the shaft to align the rotation locking portion and the fitting portion with each other for fitting them together, without the constraint of engagement between the externally threaded portion and the internally threaded portion. This structure is easier to assemble than a structure including no cylindrical portion.

Third Aspect

[0014] In the three-way valve according to the second aspect, the shaft has a groove on an outer peripheral surface of the shaft. The groove receives an O-ring sealing between the shaft and the bearing in the guide. The groove is located to have the O-ring downward from a lower end of the internally threaded portion with the upper end of the rotation locking portion located downward from the upper end of the internally threaded portion when the shaft placed through the bearing in the guide is at the uppermost position to which the shaft is movable in the axial direction.

[0015] In the three-way valve according to the third aspect, the O-ring fixed to the shaft eliminates a backup ring or a retaining ring for fixing the O-ring to the bearing in the guide.

[0016] The three-way valve thus includes fewer components and is easier to assemble. Although the O-ring moves in the axial direction together with the shaft, the O-ring does not reach the internally threaded portion when the shaft is at the uppermost position to which the shaft is movable. The O-ring is thus free from damage caused by contact with the boundary between the bearing and the internally threaded portion or by contact with the internal thread.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a diagram of a water heater 1 including a three-way valve 26 according to an embodiment, showing an example overall structure.

[0018] FIG. 2 is a cross-sectional view of the three-way valve 26 according to the present embodiment, showing its internal structure.

[0019] FIG. 3 is an exploded perspective view of the three-way valve 26 according to the present embodiment.

[0020] FIGS. 4A, 4B, 4C, and 4D are diagrams of a driver 50 including a cylindrical portion 54 and a driver 50 including no cylindrical portion 54, comparing the likelihood of grease adhering to the distal end of a small-diameter portion 35a of a shaft 35.

[0021] FIGS. 5A, 5B, and 5C are diagrams describing an assembly sequence of the three-way valve 26 in which the shaft 35 in a shaft valve element 36 is inserted through a bearing 42 in a guide 41, and then an externally threaded portion 53 on the driver 50 is screwed into an internally threaded portion 43 on the guide 41.

DETAILED DESCRIPTION

[0022] FIG. 1 is a diagram of a water heater 1 including a three-way valve 26 according to an embodiment, showing its overall structure. As illustrated, the water heater 1 includes a burner 3 in a

combustion chamber **2** to burn fuel gas. The fuel gas is supplied through a gas pipe **4** including a main valve **5** that opens and closes the gas pipe **4** and a proportioning valve **6** that regulates the flow rate of fuel gas. The water heater **1** further includes a combustion fan **10** that supplies combustion air to the burner **3** from below, a spark plug **11** that ignites the burner **3** with a spark discharge, and a flame rod **12** that detects a flame (ignition) of the burner **3**. The speed of the combustion fan **10** is controlled based on the degree of opening of the proportioning valve **6** to adjust the air-fuel ratio to a predetermined ratio.

[0023] The water heater **1** includes a heat exchanger **13** above the burner **3** and an exhaust port **14** at the top of the water heater **1**. Exhaust gas generated by combustion in the burner **3** flows upward with the combustion fan **10** blowing and flows through the heat exchanger **13** and the exhaust port **14** to outside the water heater **1**. The heat exchanger **13** has an upstream end connected to a water pipe **20** for supplying clean water, and a downstream end connected to a hot-water pipe **21**. In the water heater **1** according to the present embodiment, the water pipe **20** is connected to the hot-water pipe **21** with a bypass pipe **22**. A portion of the water pipe **20** upstream from the connection with the bypass pipe **22** may be referred to as an upstream water pipe **20a**. A portion of the water pipe **20** downstream from the connection with the bypass pipe **22** may be referred to as a downstream water pipe **20b**.

[0024] The upstream water pipe **20a** includes a water temperature sensor **23** that detects the temperature of supplied clean water, a flow sensor **24** that detects the flow rate of clean water, and a water flow servo **25** that opens and closes the upstream water pipe **20a** and controls the flow rate. These components are located in this order from upstream. The three-way valve **26** is located at the connection between the water pipe **20** and the bypass pipe **22**. The three-way valve **26** according to the present embodiment is used as a flow divider valve for dividing a flow of clean water supplied to the water heater **1** into a flow to the heat exchanger **13** and a flow to the bypass pipe **22** without being through the heat exchanger **13**. The three-way valve **26** can change the amount of water flowing into the heat exchanger **13** and the amount of water flowing into the bypass pipe **22**. The details of the three-way valve **26** will be described later with reference to another figure.

[0025] The clean water supplied to the heat exchanger **13** is heated by heat exchange with

[0026] exhaust gas from the burner **3** using the heat exchanger **13**. The resulting hot water flows to the hot-water pipe **21**. The hot water flowing to the hot-water pipe **21** mixes with the clean water passing through the bypass pipe **22** and is supplied to a bathtub or a hot water tap (not shown). The three-way valve **26** can change the mixing ratio between the water heated by the heat exchanger **13** and the clean water passing through the bypass pipe **22**.

[0027] The hot-water pipe **21** includes a heated water temperature sensor **27** that detects the temperature of hot water immediately after flowing out of the heat exchanger **13**, and an outlet water temperature sensor **28** that detects the temperature (outlet water temperature) of hot water after mixing with clean water downstream from the connection with the bypass pipe **22**. The temperature detected by the outlet water temperature sensor **28** is lower than the temperature detected by the heated water temperature sensor **27**. The three-way valve **26** regulates the mixing ratio between the water heated by the heat exchanger **13** and the clean water passing through the bypass pipe **22** to reduce fluctuations in the outlet water temperature (the temperature of hot water supplied to the bathtub or the hot water tap).

[0028] FIG. **2** is a cross-sectional view of the three-way valve **26** according to the present embodiment, showing its internal structure. As illustrated, the three-way valve **26** includes a casing **30** containing a straight flow path **31**. The straight flow path **31** has an end continuous with a first port **32a**, the other end continuous with a second port **32b**, and an intermediate portion continuous with a third port **32c**. In the water heater **1** according to the present embodiment, the first port **32a** is connected to the downstream water pipe **20b**, the second port **32b** is connected to the bypass pipe **22**, and the third port **32c** is connected to the upstream water pipe **20a** (refer to FIG. **1**).

[0029] The straight flow path **31** includes a substantially circular first valve orifice **33a** between the

first port **32a** and the third port **32c**. The first valve orifice **33a** has an opening area changeable in response to a first valve element **34a** moving toward or away from the first valve orifice **33a**. The first valve element **34a** is tapered toward the first valve orifice **33a**. Similarly, the straight flow path **31** includes a substantially circular second valve orifice **33b** between the second port **32b** and the third port **32c**. The second valve orifice **33b** has an opening area changeable in response to a second valve element **34b** moving toward or away from the second valve orifice **33b**. The second valve element **34b** is tapered toward the second valve orifice **33b**. The first valve element **34a** and the second valve element **34b** are located at an end of a shaft **35**. The shaft **35** is drivable by a motor (not shown) to move in the axial direction parallel to the straight flow path **31**. In the example described below, as shown in FIG. 2, the shaft **35** is oriented with its axial direction aligned with the up-down direction (vertical direction). The direction in which the first valve element **34a** and the second valve element **34b** are located is downward, and the opposite direction is upward. However, the three-way valve **26** may be oriented differently and may be used with the axial direction of the shaft **35** aligned with the horizontal direction.

[0030] The first valve element **34a** and the second valve element **34b** moving along the straight flow path **31** together with the shaft **35** can change the amount of fluid (water in the present embodiment) flowing between the first port **32a** and the third port **32c** and the amount of fluid flowing between the second port **32b** and the third port **32c**. When the shaft **35** moves upward, for example, the first valve element **34a** moves toward the first valve orifice **33a** and decreases the opening area of the first valve orifice **33a**, thus decreasing the amount of water flowing between the first port **32a** and the third port **32c**. At the same time, the second valve element **34b** moves away from the second valve orifice **33b** and increases the opening area of the second valve orifice **33b**, thus increasing the amount of water flowing between the second port **32b** and the third port **32c**. In contrast, when the shaft **35** moves downward, the first valve element **34a** moves away from the first valve orifice **33a** and increases the opening area of the first valve orifice **33a**, thus increasing the amount of water flowing between the first port **32a** and the third port **32c**. At the same time, the second valve element **34b** moves toward the second valve orifice **33b** and decreases the opening area of the second valve orifice **33b**, thus decreasing the amount of water flowing between the second port **32b** and the third port **32c**.

[0031] The casing **30** incorporates, in its upper portion, an inner member **40** with a guide **41** (described later). The shaft **35** is placed through a bearing **42** in the guide **41**. The shaft **35** is supported by the bearing **42** and is movable in the axial direction (the up-down direction in the figure). The distal end (the upper end in the figure) of the shaft **35** opposite to the first valve element **34a** and the second valve element **34b** is received inside a cylindrical driver **50** and bonded to the driver **50** by heat welding. The driver **50** includes, on its outer peripheral surface, a gear **51** (described later) to which a driving force is transmitted from the motor.

[0032] As illustrated, a cover plate **60** covers an upper portion of the casing **30** incorporating the inner member **40**. The cover plate **60** has a through-hole **60a** through which an upper portion of the guide **41** protrudes upward. A motor case **61** is mounted above the cover plate **60**. The motor case **61** accommodates the motor and a device (not shown) that transmits a driving force from the motor. The motor case **61** also accommodates the driver **50** bonded to the distal end of the shaft **35**.

[0033] FIG. 3 is an exploded perspective view of the three-way valve **26** according to the present embodiment. FIG. 3 does not show the casing **30**, the cover plate **60**, or the motor case **61**. In the three-way valve **26** according to the present embodiment, the first valve element **34a**, the second valve element **34b**, and the shaft **35** are an integral piece as a shaft valve element **36**. The first valve element **34a** and the second valve element **34b** are located at an end (the lower end in the figure) of the shaft **35**, as described above.

[0034] As illustrated, the shaft **35** has a nonuniform diameter and includes three portions

[0035] with different diameters, or specifically, a small-diameter portion **35a**, a medium-diameter portion **35b**, and a large-diameter portion **35c**. The small-diameter portion **35a** with the smallest

diameter is at the top and bonded to the driver **50**. The large-diameter portion **35c** with the largest diameter is at the bottom and includes the valve elements **34a** and **34b**. The medium-diameter portion **35b** with a medium diameter is located between the small-diameter portion **35a** and the large-diameter portion **35c**. The small-diameter portion **35a** and the large-diameter portion **35c** are substantially circular in a cross section perpendicular to the axial direction of the shaft **35**, whereas the medium-diameter portion **35b** has a rotation locking shape being a noncircle in a cross section. In the present embodiment, the rotation locking shape is an I shape, which is the shape of a cylinder with its cylindrical surface cut away, in the axial direction, along two planes parallel to each other. The rotation locking shape of the medium-diameter portion **35b** is not limited to the I shape and may be a D shape, which is the shape of a cylinder with its cylindrical surface cut away, in the axial direction, along a single plane. The medium-diameter portion **35b** in the present embodiment corresponds to a rotation locking portion in one or more aspects of the present invention.

[0036] The large-diameter portion **35c** has, on its outer peripheral surface, two grooves **35d** to receive O-rings **37**. The grooves **35d** extend in the circumferential direction and are spaced from each other in the axial direction. The O-rings **37** can seal between the outer peripheral surface of the large-diameter portion **35c** and the inner peripheral surface of the bearing **42** when the shaft **35** is placed through the bearing **42** in the guide **41**.

[0037] The shaft valve element **36** in the present embodiment further includes multiple (three in the present embodiment) guiding portions **38** protruding downward (opposite to the driver **50**) from the second valve element **34b**. The guiding portions **38** are substantially equally spaced in the circumferential direction. The multiple guiding portions **38** are placed through the second valve orifice **33b** to reduce wobbles of the shaft valve element **36** with respect to the axial direction of the shaft **35**. This allows more accurate movement of the first valve element **34a** and the second valve element **34b**.

[0038] The shaft valve element **36** in the present embodiment includes multiple (four in the present embodiment) protrusions **39** protruding upward (toward the driver **50**) from the first valve element **34a**. The protrusions **39** are substantially equally spaced on the peripheral surface of the large-diameter portion **35c**. As described later, the multiple protrusions **39** come in contact with the lower end (the end facing the first valve element **34a**) of the bearing **42** in the guide **41** and define the uppermost position to which the shaft **35** is movable in the axial direction (the limit of movement toward the driver **50**).

[0039] The inner member **40** in the present embodiment is an integral piece including the guide **41** with the bearing **42**, a valve seat **45** with the first valve orifice **33a**, and a connection portion **46** between the guide **41** and the valve seat **45**. FIG. 3 shows the inner member **40** cutaway along a plane including the centerline of the bearing **42**, allowing the inside of the bearing **42** to be visible. The shaft **35** in the shaft valve element **36** is placed through the first valve orifice **33a** in the valve seat **45** and through the bearing **42** in the guide **41**.

[0040] The bearing **42** is substantially cylindrical. The O-rings **37** seal between the outer peripheral surface of the large-diameter portion **35c** and the inner peripheral surface of the bearing **42** when the shaft **35** is placed through the bearing **42** as described above. In other words, the O-rings **37** received in the grooves **35d** on the outer peripheral surface of the large-diameter portion **35c** move inside the bearing **42** together with the shaft **35**. The O-rings **37** fixed to the shaft **35** in this manner eliminate a backup ring or a retaining ring for fixing the O-rings **37** to the bearing **42** in the guide **41**. The three-way valve **26** thus includes fewer components and is easier to assemble.

[0041] As described above, the protrusions **39** on the shaft valve element **36** come in contact with the lower end of the bearing **42** to restrict upward movement of the shaft valve element **36**. As illustrated, the guide **41** includes an internally threaded portion **43** upward from the bearing **42** (in a portion facing the driver **50**). The internally threaded portion **43** has an inner peripheral surface with an internal thread.

[0042] The driver **50** in the present embodiment is cylindrical and includes, on its outer peripheral

surface, the gear **51** to which a driving force is transmitted from the motor. The driver **50** has an insertion hole **52** inside the driver **50** to receive the small-diameter portion **35a** of the shaft **35**. The insertion hole **52** includes a fitting portion **52a** adjacent to its lower end (the end facing the first valve element **34a**). The fitting portion **52a** has a shape corresponding to the rotation locking shape of the medium-diameter portion **35b** to fit on the medium-diameter portion **35b**. The fitting portion **52a** fitting on the medium-diameter portion **35b** locks rotation of the driver **50** relative to the shaft **35**. The insertion hole **52** includes a portion adjacent to its upper end with a substantially circular shape corresponding to the shape of the small-diameter portion **35a** in a cross section perpendicular to the axial direction. The insertion hole **52** in the present embodiment extends in the up-down direction through the driver **50**. The shaft **35** is received in the insertion hole **52** through the fitting portion **52a**, with the distal end of the small-diameter portion **35a** protruding from the end of the insertion hole **52** opposite to the fitting portion **52a**. In this state, the shaft **35** and the driver **50** are bonded together by heat welding using infrared rays.

[0043] The driver **50** also includes, on its outer peripheral surface, an externally threaded portion **53** having an external thread downward from the gear **51**. The externally threaded portion **53** corresponds to the internally threaded portion **43** on the guide **41**. The driver **50** further includes a cylindrical portion **54** protruding downward from the lower end face of the externally threaded portion **53**. The cylindrical portion **54** has an outer diameter smaller than the inner diameter of the internally threaded portion **43**. The cylindrical portion **54** includes, inside the cylindrical portion **54**, the fitting portion **52a** as a part of the insertion hole **52**.

[0044] The driver **50** is driven by the motor and rotates about the axis of the shaft **35**, with the externally threaded portion **53** on the driver **50** placed in the internally threaded portion **43** on the guide **41**. The shaft **35** thus moves in the axial direction together with the driver **50** relative to the guide **41** in response to the externally threaded portion **53** being screwed into the internally threaded portion **43**.

[0045] In the three-way valve **26** described above, grease is typically applied to the externally threaded portion **53** on the driver **50** during assembly to allow the externally threaded portion **53** on the driver **50** to be smoothly screwed into the internally threaded portion **43** on the guide **41**. The grease may adhere to the distal end of the small-diameter portion **35a** of the shaft **35** inserted into the insertion hole **52** in the driver **50**. The grease placed between the driver **50** and the shaft **35** may decrease their bonding strength. In the three-way valve **26** according to the present embodiment, the driver **50** including the cylindrical portion **54** reduces the likelihood that the grease applied to the externally threaded portion **53** on the driver **50** adheres to the distal end of the small-diameter portion **35a** of the shaft **35**. This effect will now be described using a driver **50** including no cylindrical portion **54** for comparison.

[0046] FIGS. **4A** to **4D** are diagrams of the driver **50** including the cylindrical portion **54** and the driver **50** including no cylindrical portion **54**, comparing the likelihood of grease adhering to the distal end of the small-diameter portion **35a** of the shaft **35**. FIGS. **4A** to **4D** each show a cross section of the driver **50** and other components taken along a plane including the centerline of the insertion hole **52**. FIGS. **4A** and **4B** show the driver **50** including no cylindrical portion **54**. As shown in FIG. **4A**, grease **G** applied to the externally threaded portion **53** on the driver **50** may overflow from the outer peripheral surface of the externally threaded portion **53** to the lower end face of the driver **50** (the end face facing the first valve element **34a**). With the lower end face having the insertion hole **52** (fitting portion **52a**), the grease **G** may adhere to a portion around the insertion hole **52**.

[0047] As shown in FIG. **4B** in an enlarged manner, when the externally threaded portion **53** on the driver **50** with the grease **G** is screwed into and reaches the bottom of the internally threaded portion **43** on the guide **41**, the grease **G** overflows from between the externally threaded portion **53** and the internally threaded portion **43** and is squeezed out on the lower end face of the driver **50**. The grease **G** adheres to the portion around the insertion hole **52** (fitting portion **52a**) that is open in

the same lower end face. The grease G on the portion around the insertion hole 52 may adhere to the distal end of the small-diameter portion 35a of the shaft 35 inserted into the insertion hole 52 in the driver 50 after being inserted through the bearing 42 in the guide 41.

[0048] FIGS. 4C and 4D show the driver 50 including the cylindrical portion 54. As shown in FIG. 4C, the grease G applied to the externally threaded portion 53 on the driver 50 may overflow from the outer peripheral surface of the externally threaded portion 53. However, with the cylindrical portion 54 protruding downward from the lower end of the externally threaded portion 53, the grease G is less likely to adhere to the lower end face (the end face facing the first valve element 34a) of the cylindrical portion 54 in which the insertion hole 52 (fitting portion 52a) is open, although the grease G may adhere to the outer peripheral surface of the cylindrical portion 54.

[0049] As shown in FIG. 4D in an enlarged manner, when the externally threaded portion 53 on the driver 50 with the grease G is screwed into the bottom of the internally threaded portion 43 on the guide 41, the grease G may overflow from between the externally threaded portion 53 and the internally threaded portion 43. However, the overflow of the grease G is limited to the space between the outer peripheral surface of the cylindrical portion 54 and the inner peripheral surface of the bearing 42. The grease G is thus less likely to adhere to the lower end face of the cylindrical portion 54 in which the insertion hole 52 (fitting portion 52a) is open.

[0050] In the driver 50 including the cylindrical portion 54, the grease G applied to the externally threaded portion 53 on the driver 50 is less likely to adhere to the portion around the insertion hole 52 (fitting portion 52a) that is open in the lower end face of the driver 50 (cylindrical portion 54) than in the driver 50 including no cylindrical portion 54. The grease G is thus less likely to adhere to the distal end of the small-diameter portion 35a of the shaft 35 inserted into the insertion hole 52 in the driver 50. This can avoid a decrease in the bonding strength between the driver 50 and the shaft 35 caused by the grease G.

[0051] In the example shown in FIGS. 4A to 4D, the externally threaded portion 53 on the driver 50 is screwed into the internally threaded portion 43 on the guide 41, and then the small-diameter portion 35a of the shaft 35 is inserted through the bearing 42 in the guide 41 and into the insertion hole 52 in the driver 50. However, the three-way valve 26 according to the present embodiment may be assembled in a different sequence. For example, the shaft 35 in the shaft valve element 36 may be inserted through the bearing 42 in the guide 41, and then the externally threaded portion 53 may be screwed into the internally threaded portion 43 on the guide 41 while the small-diameter portion 35a of the shaft 35 is being inserted into the insertion hole 52 in the driver 50. In this case, the three-way valve 26 including the cylindrical portion 54 in the driver 50 is easier to assemble, in addition to reducing the likelihood that the grease G applied to the externally threaded portion 53 on the driver 50 adheres to the distal end of the small-diameter portion 35a of the shaft 35 as described above. This effect will now be described.

[0052] FIGS. 5A to 5C are diagrams describing an assembly sequence of the three-way valve 26 in which the shaft 35 in the shaft valve element 36 is inserted through the bearing 42 in the guide 41, and then the externally threaded portion 53 on the driver 50 is screwed into the internally threaded portion 43 on the guide 41. FIGS. 5A to 5C each show a cross section of the shaft valve element 36 and other components taken along a plane including the centerline of the shaft 35. FIG. 5A shows the shaft 35 in the shaft valve element 36 placed through the bearing 42 in the guide 41.

[0053] As described above, the shaft valve element 36 in the present embodiment includes the protrusions 39 that come in contact with the lower end (the end facing the first valve element 34a) of the bearing 42 and define the uppermost position to which the shaft 35 is movable in the axial direction (the limit of movement toward the driver 50). The grooves 35d (refer to FIG. 3) receiving the O-rings 37 are located to have the O-rings 37 downward from the lower end of the internally threaded portion 43 when the shaft valve element 36 is at the uppermost position to which the shaft valve element 36 is movable. Although the O-rings 37 move in the axial direction together with the shaft 35, the O-rings 37 do not reach the internally threaded portion 43. The O-rings 37 are thus

free from damage caused by contact with the boundary between the bearing **42** and the internally threaded portion **43** or by contact with the internal thread.

[0054] In the shaft valve element **36** in the present embodiment, when the shaft **35** is at the uppermost position to which the shaft **35** is movable in the axial direction, the small-diameter portion **35a** at the top protrudes upward from the internally threaded portion **43**, whereas the upper end of the medium-diameter portion **35b** is located downward from the upper end of the internally threaded portion **43** (inside the internally threaded portion **43**) without protruding upward from the internally threaded portion **43**, as shown in FIG. 5A. To join the driver **50**, the externally threaded portion **53** is screwed into the internally threaded portion **43** on the guide **41** while the small-diameter portion **35a** of the shaft **35** protruding from the internally threaded portion **43** is being inserted into the insertion hole **52** in the driver **50**.

[0055] In the driver **50** including no cylindrical portion **54**, the grease G on the portion around the insertion hole **52** (fitting portion **52a**) that is open in the lower end face of the driver **50** may adhere to the distal end of the small-diameter portion **35a**, as described above (refer to FIG. 4A). Further, as shown in FIG. 5B, the medium-diameter portion **35b** having the rotation locking shape is fitted into the fitting portion **52a** after the externally threaded portion **53** on the driver **50** starts being screwed into the internally threaded portion **43** on the guide. In other words, the medium-diameter portion **35b** is to be aligned with the fitting portion **52a** for their fitting under the constraint of engagement between the externally threaded portion **53** and the internally threaded portion **43**. This complicates the assembly.

[0056] In contrast, in the driver **50** including the cylindrical portion **54**, the grease G applied to the externally threaded portion **53** is less likely to adhere to the portion around the insertion hole **52** (fitting portion **52a**) that is open in the lower end face of the driver **50** (cylindrical portion **54**), as described above (refer to FIG. 4C). The grease G is thus less likely to adhere to the distal end of the small-diameter portion **35a**. Further, as shown in FIG. 5C, the cylindrical portion **54** in the driver **50** in the present embodiment protrudes by a length that allows the fitting portion **52a** inside the cylindrical portion **54** to fit on the medium-diameter portion **35b** before the externally threaded portion **53** on the driver **50** is screwed into the internally threaded portion **43** on the guide. The driver **50** with the cylindrical portion **54** placed inside the internally threaded portion **43** can thus be rotated freely about the axis of the shaft **35** to align the medium-diameter portion **35b** and the fitting portion **52a** with each other for fitting them together, without the constraint of engagement between the externally threaded portion **53** and the internally threaded portion **43**. This structure is easier to assemble.

[0057] The three-way valve **26** according to the present embodiment has been described. However, the present invention is not limited to the above embodiment and may be implemented in various manners without departing from the spirit and scope of the invention.

[0058] For example, in the above embodiment, the three-way valve **26** is used as a flow divider valve in the water heater **1**. However, the three-way valve **26** is not limited to a flow divider valve and may be used as a mixing valve for combining a flow of water heated by the heat exchanger **13** and a flow of clean water passing through the bypass pipe **22**. In this case, the first port **32a** may be connected to a portion of the hot-water pipe **21** upstream from the connection with the bypass pipe **22** (nearer the heat exchanger **13**), the second port **32b** may be connected to the bypass pipe **22**, and the third port **32c** may be connected to a portion of the hot-water pipe **21** downstream from the connection with the bypass pipe **22** (nearer the hot water tap). The three-way valve **26** may be used as a flow divider valve or a mixing valve to control any of various fluids, rather than being used in the water heater **1**.

[0059] In the above embodiment, the shaft **35** and the driver **50** are bonded together by heat welding with the small-diameter portion **35a** placed in the insertion hole **52**. However, the shaft **35** and the driver **50** may be bonded together with a method other than heat welding, such as press-fitting the shaft **35** into the insertion hole **52** or using adhesive. When such press-fitting or adhesive

is used to bond the shaft 35 and the driver 50, the insertion hole 52 may not extend through the driver 50 to the upper end, unlike the insertion hole 52 in the above embodiment extending in the up-down direction (the axial direction of the shaft 35) through the driver 50. For bonding with press-fitting or adhesive as well, any grease G adhering to the distal end of the shaft 35 may prevent proper bonding between the shaft 35 and the driver 50. Thus, the structure according to one or more aspects of the present invention can avoid a decrease in the bonding strength.

[0060] In the above embodiment, the shaft 35, the first valve element 34a, and the second valve element 34b are an integral piece as the shaft valve element 36. In some embodiments, the shaft 35, the first valve element 34a, and the second valve element 34b may be separate pieces that are bonded together to serve as the shaft valve element 36, rather than being an integral piece.

However, the shaft 35, the first valve element 34a, and the second valve element 34b being an integral piece as the shaft valve element 36 in the above embodiment can eliminate, for example, a stop ring or a push nut for bonding separate such components. The three-way valve 26 thus includes fewer components and is easier to assemble.

[0061] In the above embodiment, the protrusions 39 are included in the shaft valve element 36 to define the uppermost position to which the shaft valve element 36 (shaft 35) is movable in the axial direction. In some embodiments, the guide 41 may include multiple protrusions protruding downward from the lower end of the bearing 42 and substantially equally spaced in the circumferential direction. The protrusions may come in contact with the first valve element 34a to define the uppermost position to which the shaft valve element 36 is movable.

REFERENCE SIGNS LIST

[0062] 1 water heater [0063] 2 combustion chamber [0064] 3 burner [0065] 4 gas pipe [0066] 5 main valve [0067] 6 proportioning valve [0068] 10 combustion fan [0069] 11 spark plug [0070] 12 flame rod [0071] 13 heat exchanger [0072] 14 exhaust port [0073] 20 water pipe [0074] 21 hot-water pipe [0075] 22 bypass pipe [0076] 23 water temperature sensor [0077] 24 flow sensor [0078] 25 water flow servo [0079] 26 three-way valve [0080] 27 heated water temperature sensor [0081] 28 outlet water temperature sensor [0082] 30 casing [0083] 31 straight flow path [0084] 32a first port [0085] 32b second port [0086] 32c third port [0087] 33a first valve orifice [0088] 33b second valve orifice [0089] 34a first valve element [0090] 34b second valve element [0091] 35 shaft [0092] 35a small-diameter portion [0093] 35b medium-diameter portion [0094] 35c large-diameter portion [0095] 35d groove [0096] 36 shaft valve element [0097] 37 O-ring [0098] 38 guiding portion [0099] 39 protrusion [0100] 40 inner member [0101] 41 guide [0102] 42 bearing [0103] 43 internally threaded portion [0104] 45 valve seat [0105] 46 connection portion [0106] 50 driver [0107] 51 gear [0108] 52 insertion hole [0109] 52a fitting portion [0110] 53 externally threaded portion [0111] 54 cylindrical portion [0112] 60 cover plate [0113] 60a through-hole [0114] 61 motor case

Claims

1. A three-way valve, comprising: a first port, a second port, and a third port; a straight flow path including an end continuous with the first port, another end continuous with the second port, an intermediate portion continuous with the third port, a first valve orifice between the first port and the third port, the first valve orifice having an opening area changeable in response to a first valve element moving toward or away from the first valve orifice, and a second valve orifice between the second port and the third port, the second valve orifice having an opening area changeable in response to a second valve element moving toward or away from the second valve orifice; a shaft drivable by a motor to move in an axial direction of the shaft parallel to the straight flow path, the shaft including a first end at which the first valve element and the second valve element are located, the first valve element and the second valve element being movable along the straight flow path together with the shaft to change an amount of fluid flowing between the first port and the third

port and an amount of fluid flowing between the second port and the third port; a guide including a bearing through which the shaft is placed, the bearing supporting the shaft to allow the shaft to move in the axial direction; and a driver being cylindrical and including, on an outer peripheral surface of the driver, a gear to which a driving force is transmitted from the motor, the driver having an insertion hole being open in an end face of the driver facing the first valve element and the second valve element, the insertion hole receiving a second end of the shaft opposite to the first valve element and the second valve element, the driver being bonded to the second end of the shaft, wherein where a direction in which the driver is located with respect to the shaft in the axial direction is upward, and an opposite direction in which the first valve element and the second valve element are located with respect to the shaft in the axial direction is downward, the driver includes, downward from the gear, an externally threaded portion having an outer peripheral surface with an external thread, and the guide includes, upward from the bearing, an internally threaded portion having an inner peripheral surface with an internal thread, when the driver is driven by the motor and rotates about an axis of the shaft with the externally threaded portion placed in the internally threaded portion, the shaft moves in the axial direction in response to the externally threaded portion being screwed into the internally threaded portion, and the driver includes a cylindrical portion protruding downward from a lower end face of the externally threaded portion, the cylindrical portion includes the insertion hole inside the cylindrical portion, and the cylindrical portion has an outer diameter smaller than an inner diameter of the internally threaded portion.

2. The three-way valve according to claim 1, wherein the shaft includes a rotation locking portion having a rotation locking shape being a noncircle in a cross section perpendicular to the axial direction, the driver includes a fitting portion inside the cylindrical portion, the fitting portion is a part of the insertion hole, and the fitting portion has a shape corresponding to the rotation locking shape to fit on the rotation locking portion, the rotation locking portion has an upper end located downward from an upper end of the internally threaded portion when the shaft placed through the bearing in the guide is at an uppermost position to which the shaft is movable in the axial direction, and the cylindrical portion in the driver protrudes by a length allowing the fitting portion to fit on the rotation locking portion before the externally threaded portion is screwed into the internally threaded portion.

3. The three-way valve according to claim 2, wherein the shaft has a groove on an outer peripheral surface of the shaft, and the groove receives an O-ring sealing between the shaft and the bearing in the guide, and the groove is located to have the O-ring downward from a lower end of the internally threaded portion with the upper end of the rotation locking portion located downward from the upper end of the internally threaded portion when the shaft placed through the bearing in the guide is at the uppermost position to which the shaft is movable in the axial direction.
